

Faheem Shah Umer Vattam Kandathil

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WORK EXPERIENCE

Werkstudent

IAV GmbH

Sep 2024 – Present

Bavaria, Germany

- Developed three production AI applications used by an estimated 50+ users, including a standalone internal RAG assistant that retrieves project information from PDFs, OneNote, emails, Word, and Excel.
- Built a four-agent system comprising an orchestrator for decision-making and task routing, an analysis agent for numerical ECU signal analysis, a plotting agent for visualizing results, and a RAG agent for retrieving signal information from documentation and e-handbooks.
- Designed diagnostic logic for fuel tank leakage detection as part of functional safety development.
- Measured and analysed ECU signals using ETAS MDA and INCA, validating ECU safety logic through simulation, numerical signal analysis, and prototype vehicle testing for an automotive OEM.

Masterarbeit

KTmfk – Lehrstuhl für Konstruktionstechnik | FAU Erlangen Nuremberg

Apr 2026 – Present

- Developed and evaluated a LangGraph-based framework orchestrating seven specialized agents for tolerance management in product development.
- Validated the prototype on several CAD files while integrating engineering knowledge retrieval, CAD/STEP and PMI processing, Monte Carlo tolerance analysis, visualization, FE meshing, and Siemens VisVSA execution.
- Implemented RAG, structured engineering artifacts, deterministic Python calculations, traceable agent handoffs, and human-in-the-loop clarification.

Projektarbeit

KTmfk – Lehrstuhl für Konstruktionstechnik | FAU Erlangen Nuremberg

Apr 2025 – Sep 2025

Grade: 1.0

- Built a RAG knowledge base from approximately 4,000 technical documents, including books, articles, standards, project papers, and theses.
- Evaluated the prototype on 20+ engineering questions, achieving approximately 95% retrieval and answer accuracy while measuring response time and user satisfaction.

Software Engineer and Content Creator

Baabte System Technologies Pvt Ltd

Jan 2023 – Apr 2023

Calicut, India

- Built working models and created technical content for Robotics, IoT, and AI projects using Arduino, MicroPython, and Raspberry Pi.
- Tools: Jira, Arduino IDE, Fritzing, MicroPython, NodeMCU, ESP32/ESP8266.

Digital Twin Benchmarking in Electronics

Caresoft Global

Apr 2022 – Jul 2022

Coimbatore, India

- Benchmarked IC components, researched architecture, manufacturer, and production cost, and compiled a final report on an electric vehicle module for use in future upgrades.

Intern

Technologies Global Pvt. Ltd

Feb 2021 – Mar 2021

Bangalore, India

- Trained in ML algorithms and Python using TensorFlow, Keras, and Pandas in Jupyter Notebook.

EDUCATION

Master's in Electromobility ACES | FAU Erlangen Nuremberg

| Present | Current GPA: 1.6

- Specializing in AI, autonomous driving, and sustainable mobility.
- Strong knowledge of electromobility technologies, including electronics, AI, and machine learning.
- Experience with simulation and modelling of electromobility systems.
- Proficient in Python.

EDUCATION

Bachelor in Mechatronics Engineering | HICET Coimbatore | Jul 2018 – May 2022 | CGPA: 8.2/10

- Experience with Mechatronics systems, including sensors, actuators, and controllers.
- Proficient in Python, C++, MATLAB, MSC ADAMS, Arduino and Windows Office.

SKILLS

AI / Agentic Systems:	LLMs, RAG, Multi-Agent Systems, LLMOps, LangChain, LangGraph, Hugging Face, Qdrant, Embeddings, Human-in-the-Loop Workflows, MCP, A2A, Google ADK, SpaCy, NLTK, TensorFlow, Keras.
Engineering / Simulation:	Tolerance Management, CAD/STEP and PMI Processing, Tolerance Analysis, Open-Cascade/XCAF, Gmsh, Siemens VisVSA, Matplotlib, MATLAB, MSC ADAMS.
Automotive / Embedded:	Functional Safety (ECU), ETAS MDA, ETAS INCA, Digital Twin, Arduino, MicroPython, Raspberry Pi, Soldering, Welding.
Programming / Tools:	Python, C, C++, PyQt5, Streamlit, Git, GitLab, Bitbucket, Jira, Confluence, Word-Press, Excel, Word.
Soft Skills:	Leadership, Teamwork, Communication, Problem-solving, Punctuality.

PROJECTS

- **LanternLM – Offline Desktop AI Assistant**

Built a portable Windows desktop app that runs GGUF language models fully offline via llama.cpp. Implemented streamed inference, automatic model discovery, SQLite chat persistence, and a local file workspace within a context-isolated Electron architecture.

Tools: Electron, React, TypeScript, llama.cpp, GGUF, SQLite, Vite

Link: github.com/faheem-shah-umer/LanternLM

- **SITY – Agile Workspace Presence Monitor**

Built a privacy-preserving hardware-software prototype covering approximately six rooms, with real-time occupancy displayed on a web app and physical boards. Led hardware selection, schematic design, and microcontroller programming.

Tools: GitLab, Arduino IDE, Firebase, Flask, Miro

Link: mad.tf.fau.de/research/projects/sity/

- **Portable Robotic Arm (Android-Controlled)**

Built a sub-3kg robotic arm controllable via Android. Led a team of four through the design, build, and final report.

Tools: Arduino IDE/SDK, Android Development Tools, C++, Java.

- **Scholarship Web Portal**

Built a centralized website for students to browse and apply for scholarships. Led a four-member team as the sole developer and received a Certificate of Appreciation from the Department of Mechatronics, Hindustan College of Engineering and Technology.

Tools: WordPress, Ajax.

CERTIFICATIONS & TRAINING

Microsoft

Build Cross-Platform Multi-Agent Solutions Using the Agent2Agent Protocol in Microsoft Copilot Studio – [Certificate](#)

LangChain Academy

Foundation: Introduction to LangChain – Python – [Certificate](#)

Foundation: Introduction to LangGraph – Python – [Certificate](#)

Project: Ambient Agents with LangGraph – [Certificate](#)

Project: Deep Research with LangGraph – [Certificate](#)

Google

Deploy Multi-Agent Architectures – [Certificate](#)

University of Toronto / Coursera

Introduction to Self-Driving Cars (August 2020)

LANGUAGES

German: A2 level

English: IELTS band 7